



Engineering Drawings Annotations

Project Name: Engineering Drawings Segmentation.

Client: FutureCalc.Ai

Annotation Team: Zeeshan khan, Daniyal Khan, Shayan Khan

Tool used: CVAT

1. Project Overview

Annotated engineering drawing images for a computer vision dataset aimed at detecting and segmenting various type of pipes. The project involved creating precise masks annotations to generate high-quality training data for an AI model.

2. Dataset Description

- **Total images:** 500 Images
- **Total classes:** 75
- **Resolutions:** 6623 × 4678 pixels
- **Source:** provided by FutureCalc.ai
- **Annotation Type:** Mask Segmentation
 - Classes Annotated

ID	Class	ID	Class	ID	Class
1	S1-P2-110	26	VP1-S13-42/W	51	KV1-X7-32/WB
2	S1-P2-75	27	VS1-S13-42/W	52	KV1-X7-40/WB
3	S1-P2-160	28	VS1-S13-54/W	53	KV1-X7-63/WB
4	D1-E4-110	29	S2-P5-50	54	KV1-X7-50/WB
5	KV1-E13-90	30	VS1-S13-22/W	55	VP1-S1-65/WB
6	KV1-E13-75	31	VS1-S13-15/W	56	VVC1-X7-32/WB
7	wall	32	S2-P5-110	57	FJV1-S6-50/WB
8	FJV1-S6-50/W	33	VV1-X31-16	58	S2-E4-110
9	S3-R8-110	34	VS1-X31-16	59	VV1-X7-40/WB
10	S3-P2-160	35	VVC1-X7-16/W	60	KV1-X7-50/W
11	S3-R8-75	36	VV1-X7-25/W	61	VV1-X7-50/W

ID	Class	ID	Class	ID	Class
12	KV1-X31-16	37	KV1-X7-32/W	62	VV1-K5-15
13	S3-R8-160	38	VVC1-X7-25/W	63	KV2-X31-16
14	KV2 - X31	39	VV1-K15/15	64	KV1-R1-12 WALLMOUNTED
15	VV1 - X31	40	KV1-K15/15	65	VS1-R1-12
16	S2-P5-75	41	KV1-X7-25/W	66	VS1-R1-12 wallmounted
17	VV1-X7-16/W	42	VV1-X7-32/W	67	VP1-S13-22/W
18	VV1-X7-20/W	43	VVC1-X7-20/W	68	KV2-X7-16/W
19	KV1-X7-20/W	44	S2-G3-50	69	KV1-R1-15 WALLMOUNTED
20	KV1-X7-16/W	45	KV1-K5-12	70	KV1-R1-15
21	VVC1-X7-32/W	46	VV1-K5-12	71	VV1-R1-15 WALLMOUNTED
22	VV1-X7-40/W	47	VS1-S13-28/W	72	KV2-X7-25/W
23	KV1-X7-40/W	48	VS1-S13- 54/WB	73	S3-P5-110
24	VS1-S13-35/W	49	VP1-S13-54/W	74	KV2-X7-32/W
25	VS1-S13-12/W	50	S2-P5-160	75	VV1-R1-12 WALLMOUNTED

1. Responsibilities

- Reviewed and organized the image dataset before annotation.
- Created accurate mask annotations for all kitchen objects.
- Ensured precise object boundaries for segmentation tasks.
- Maintained consistency across all class labels.
- Performed quality checks and corrected annotation errors.
- Delivered completed annotations according to client requirements.
- Challenges
- Annotating overlapping kitchen appliances and furniture.
- Maintaining consistent labeling across multiple object categories.
- Creating detailed masks for complex object shapes.

2. Outcome

Successfully delivered a fully annotated segmentation dataset consisting of 500 images with 75 pipes classes. The dataset was prepared for use in computer vision model training and object segmentation tasks.

3. **Skills Demonstrated**

Masks Segmentation

Image Annotation

Data Labeling

Quality Assurance

CVAT Annotation

Computer Vision Data Preparation

4. **Final output:**

Completed **500 engineering drawings annotations** with 75 **object classes** using polygon segmentation while maintaining annotation quality and consistency throughout the dataset.

Example Screenshots

Example: 1

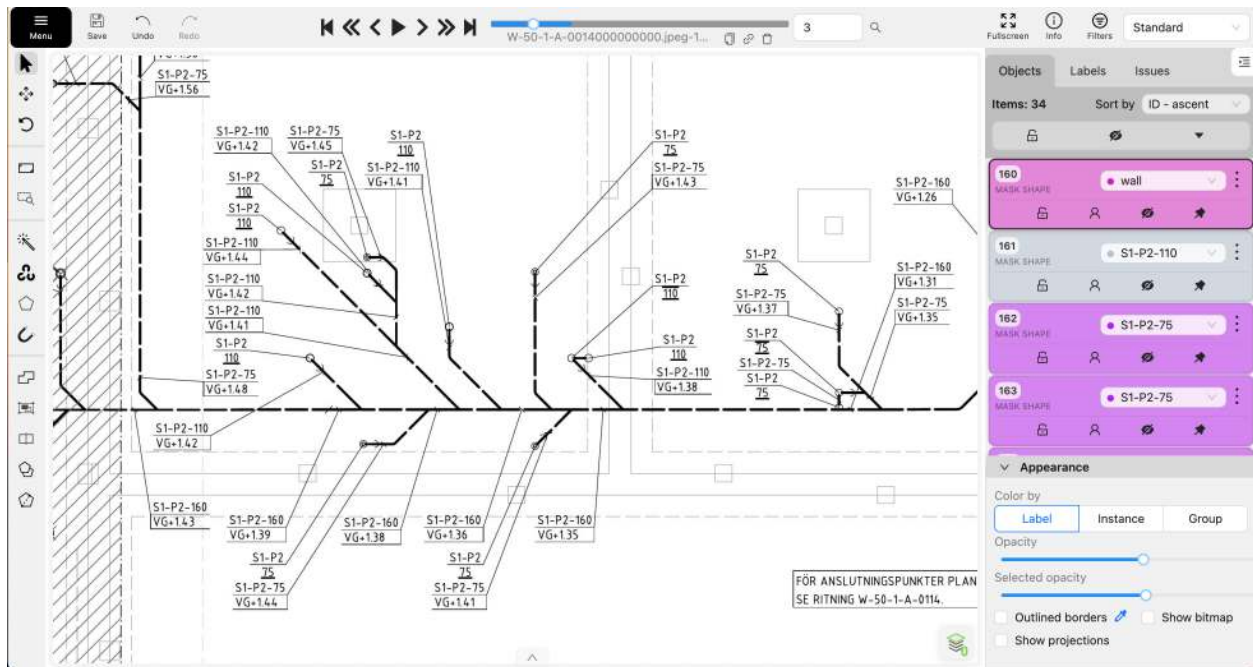


Figure 1.1 [A zoomed engineering drawing before annotation]

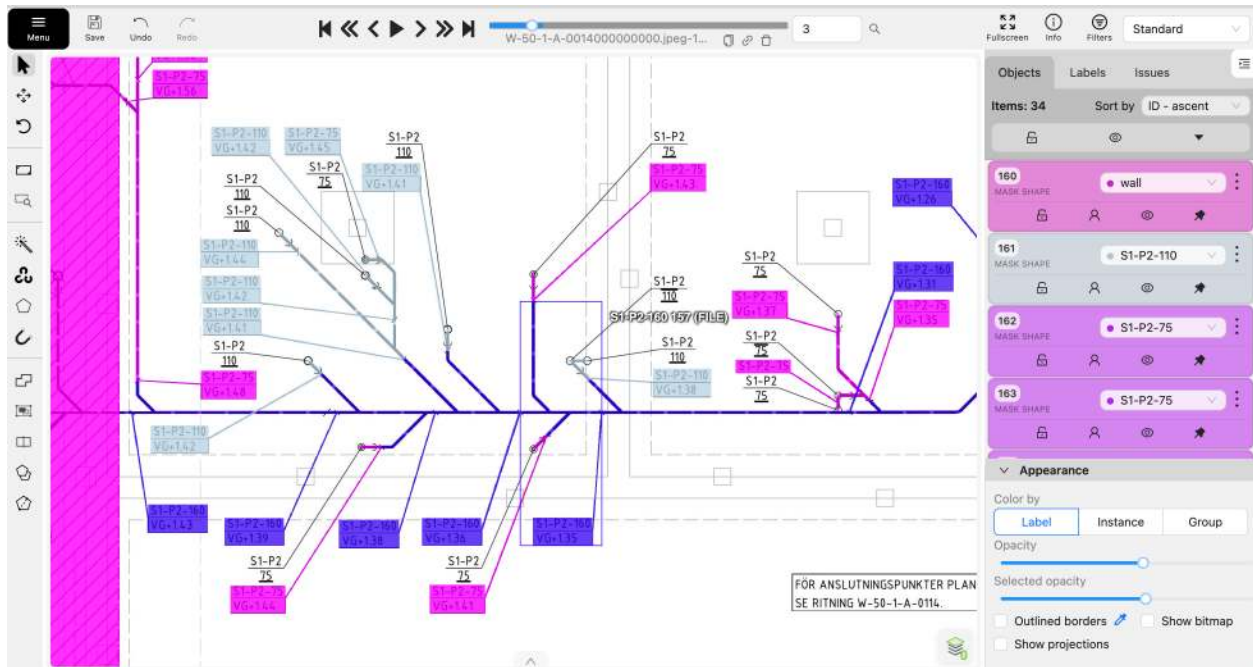


Figure 1.2 [A zoomed engineering drawing after annotation]

Example: 2

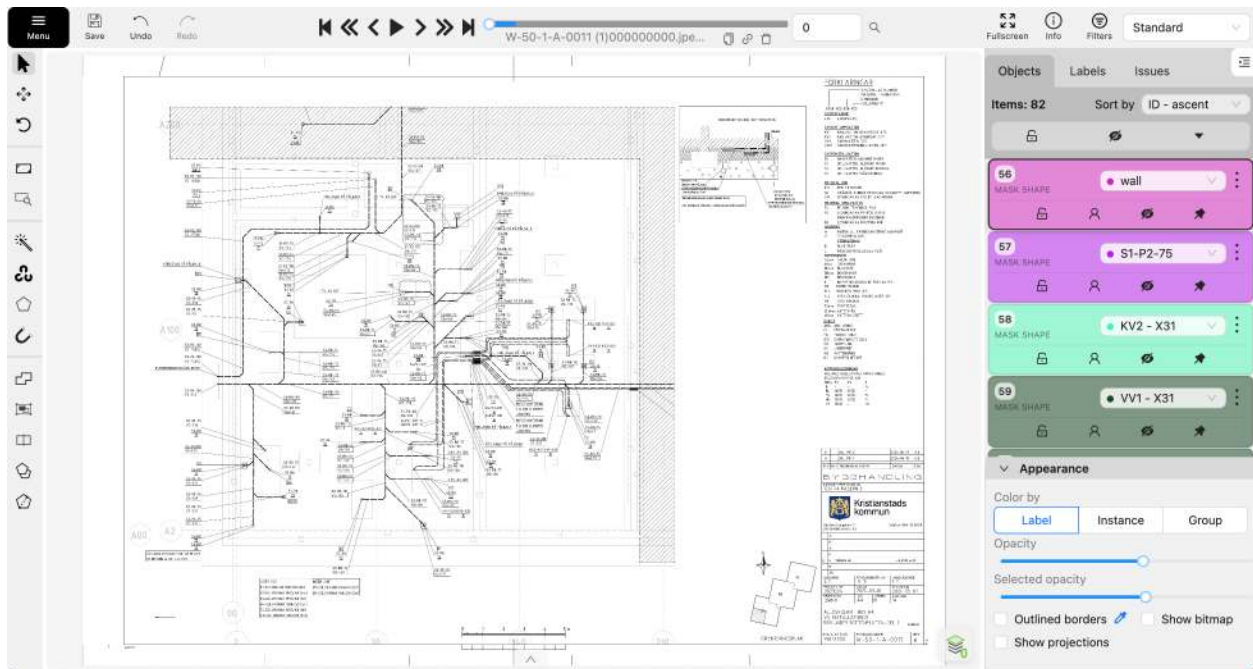


Figure 2.1 [An engineering drawing before annotation]

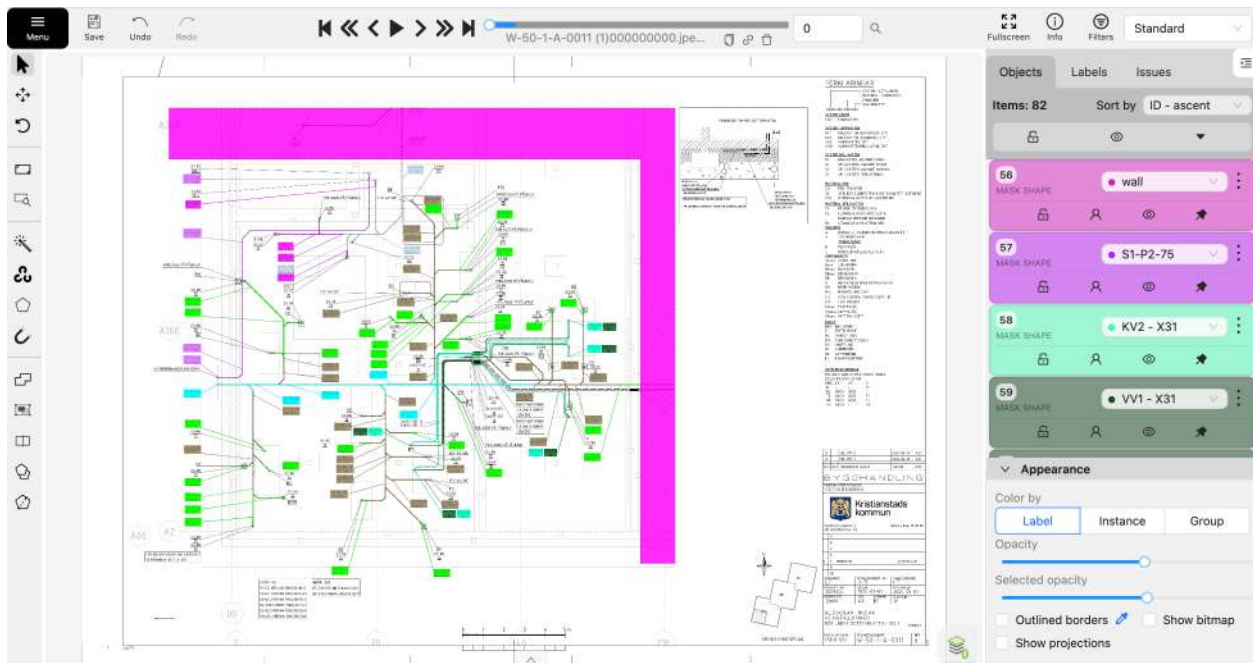


Figure 2.2 [An engineering drawing after annotation]

Example: 3

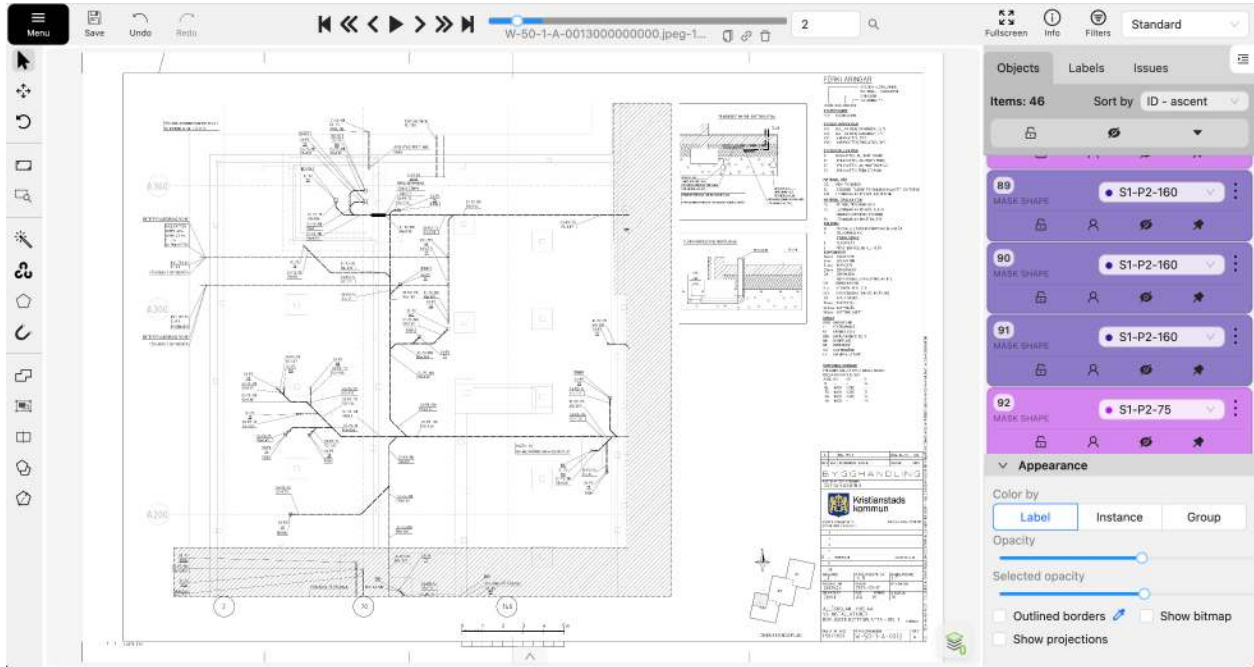


Figure 3.1 [An engineering drawing before annotation]

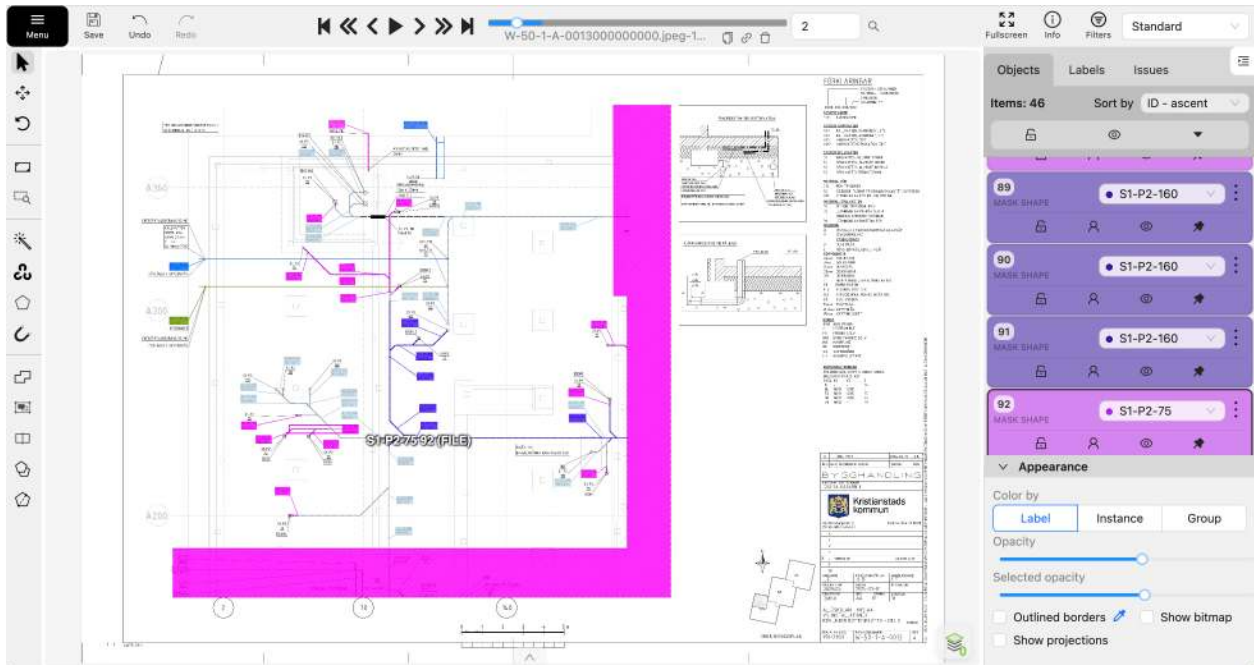


Figure 3.2 [An engineering drawing after annotation]

Example: 4

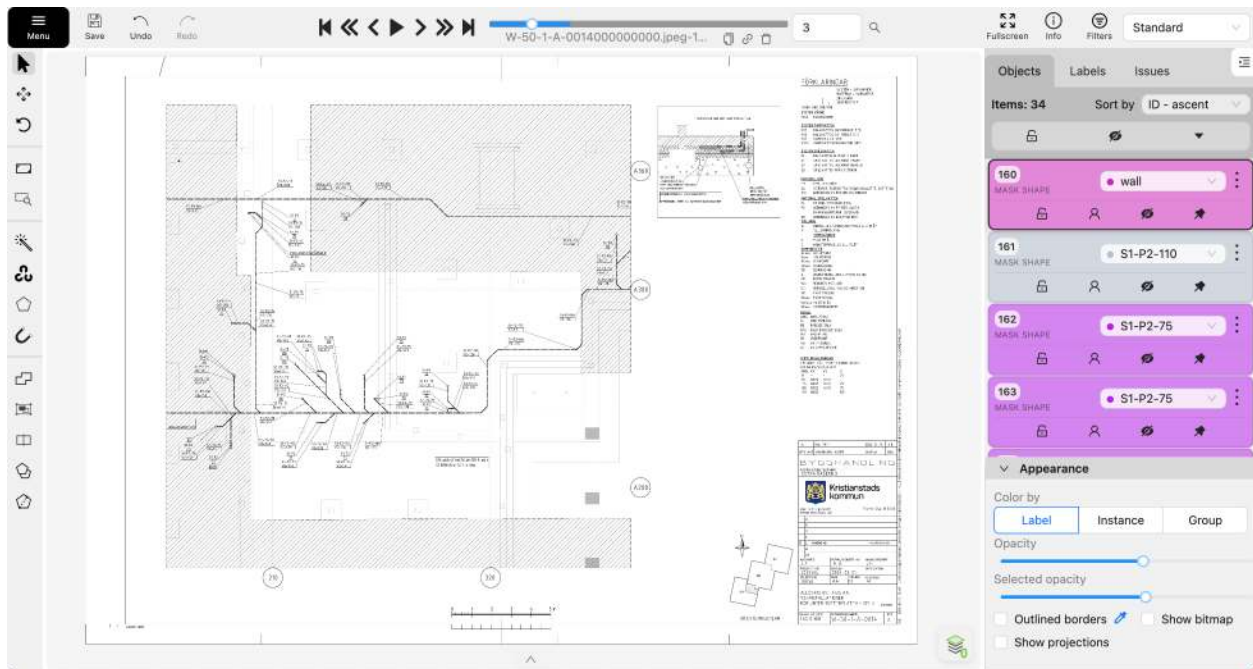


Figure 4.1 [An engineering drawing before annotation]

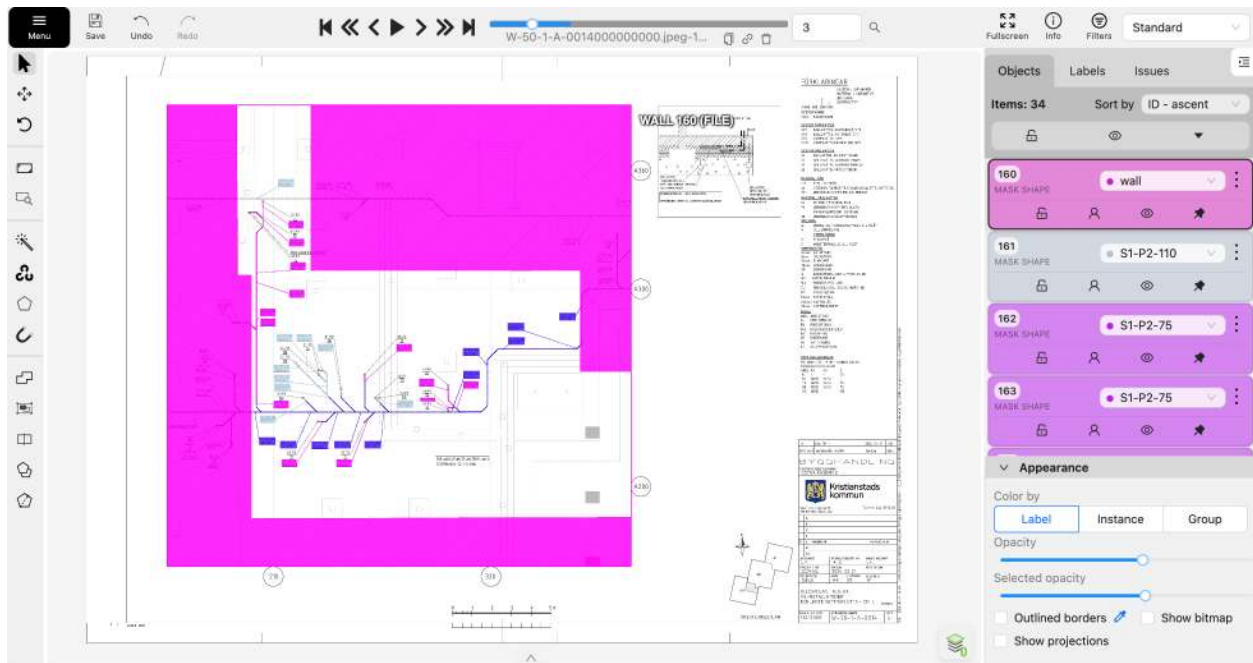


Figure 4.2 [An engineering drawing after annotation]

Example: 5

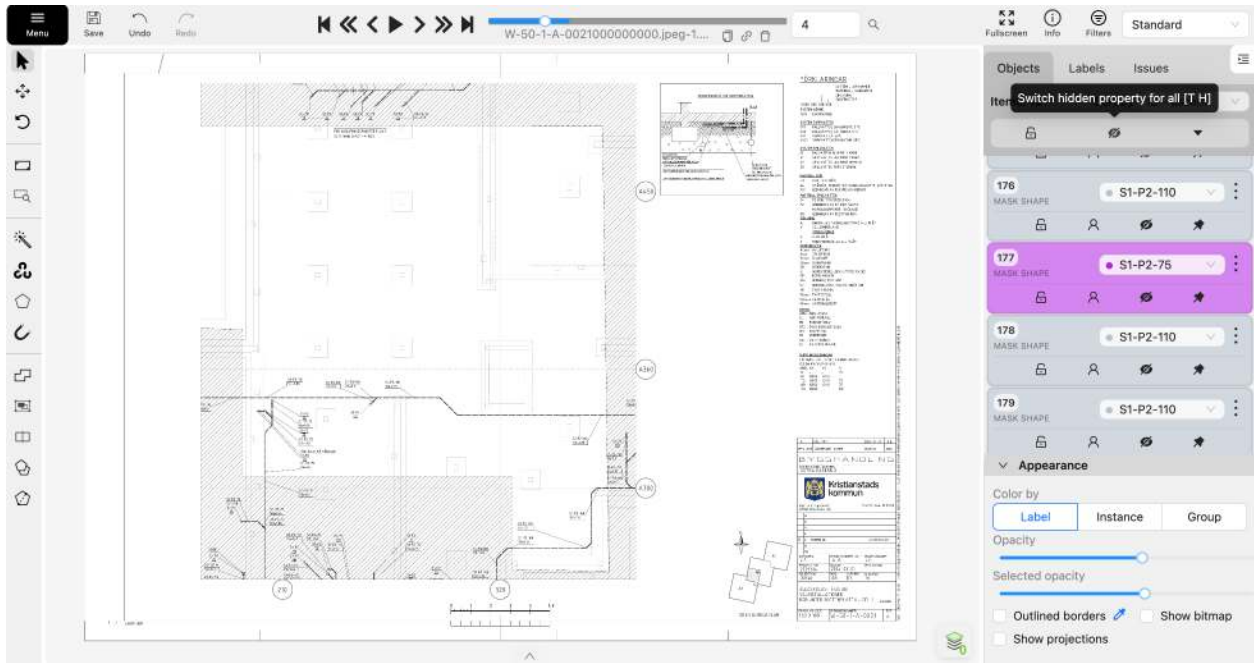


Figure 5.1 [An engineering drawing before annotation]

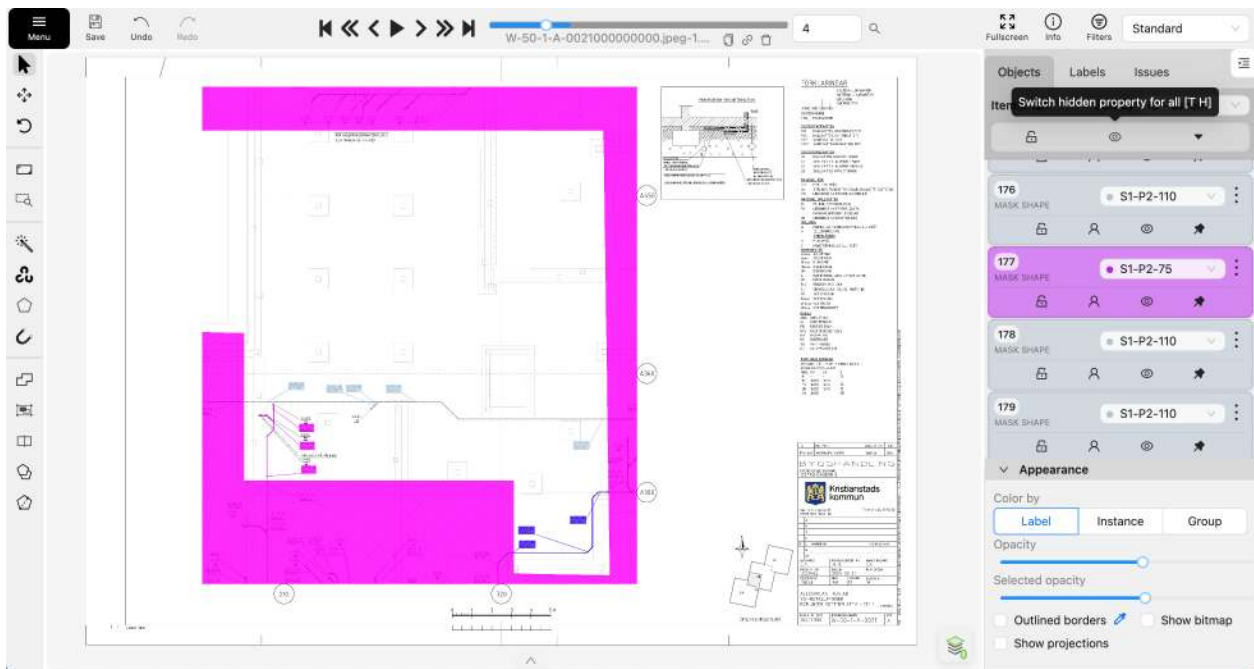


Figure 5.2 [An engineering drawing after annotation]

Example: 6

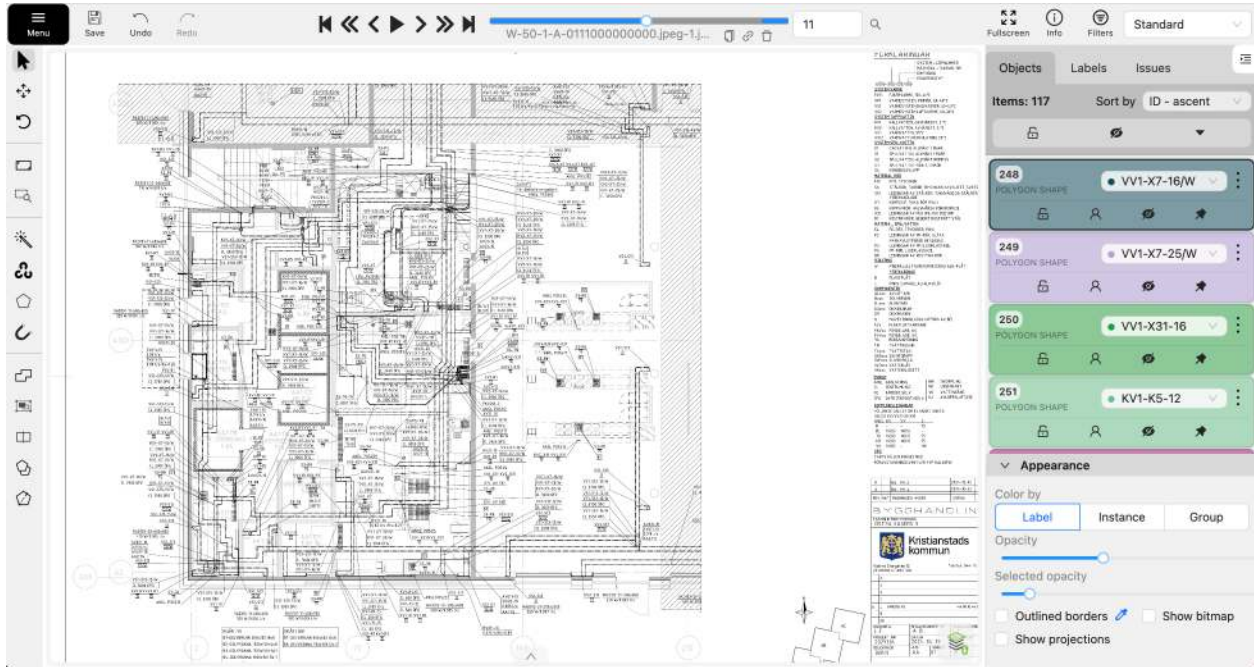


Figure 6.1 [An engineering drawing before annotation]

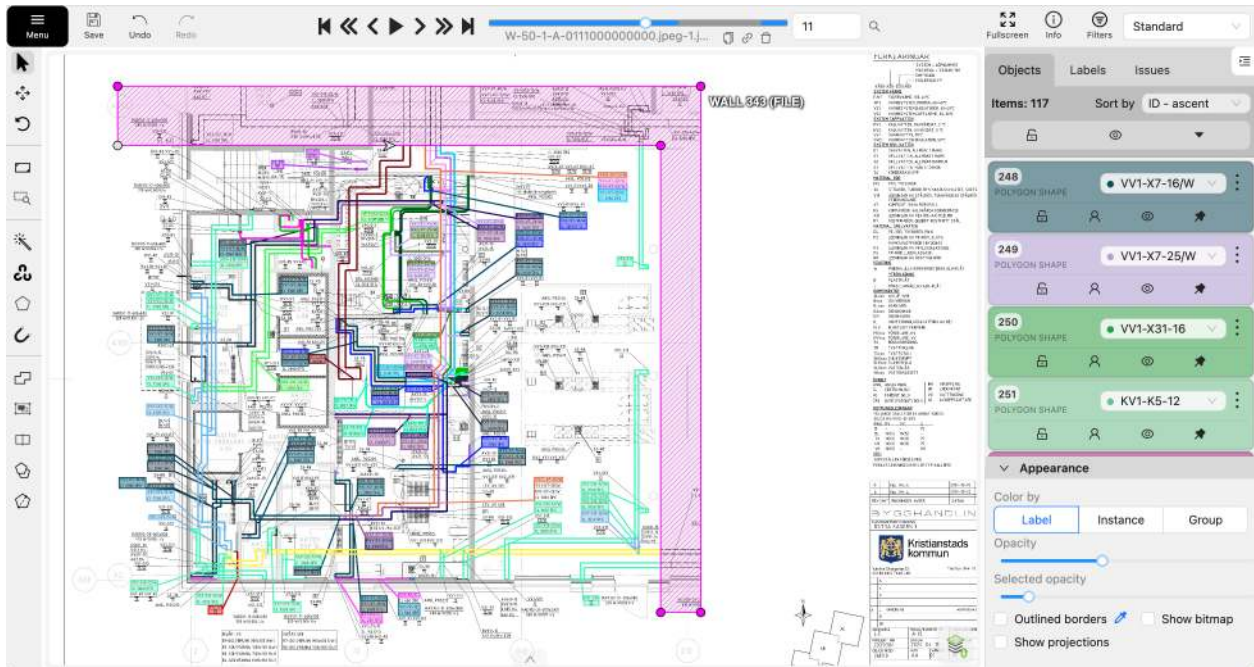


Figure 6.2 [An engineering drawing after annotation]

Example: 7

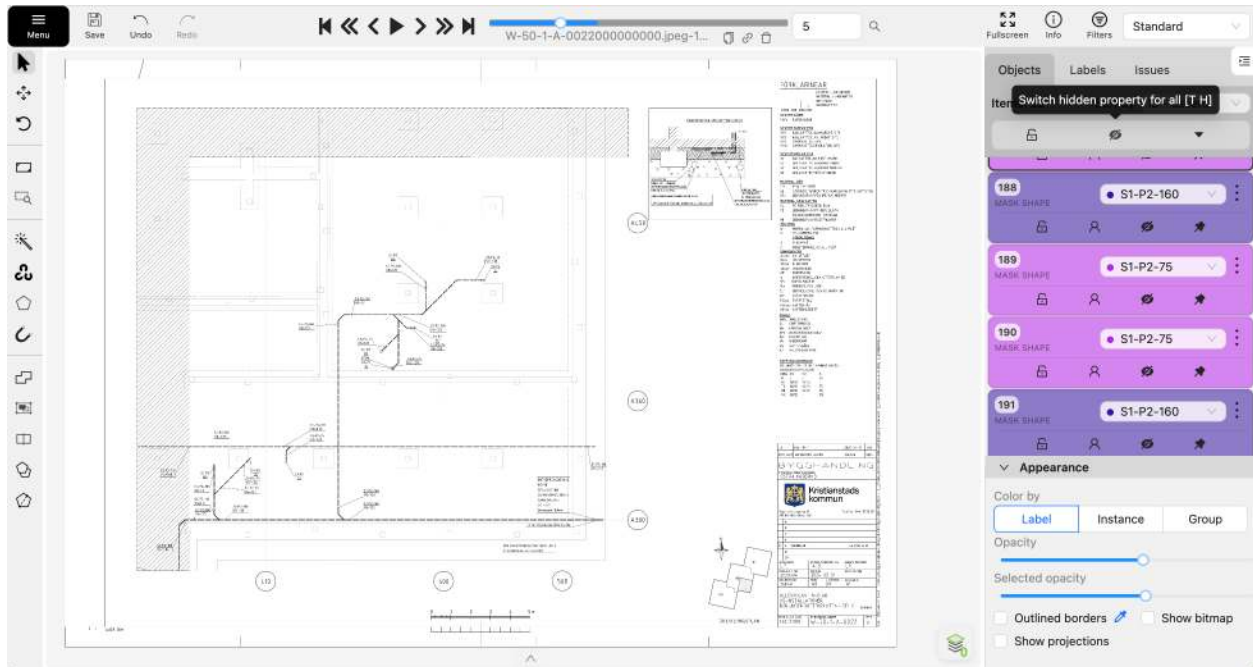


Figure 7.1 [An engineering drawing before annotation]

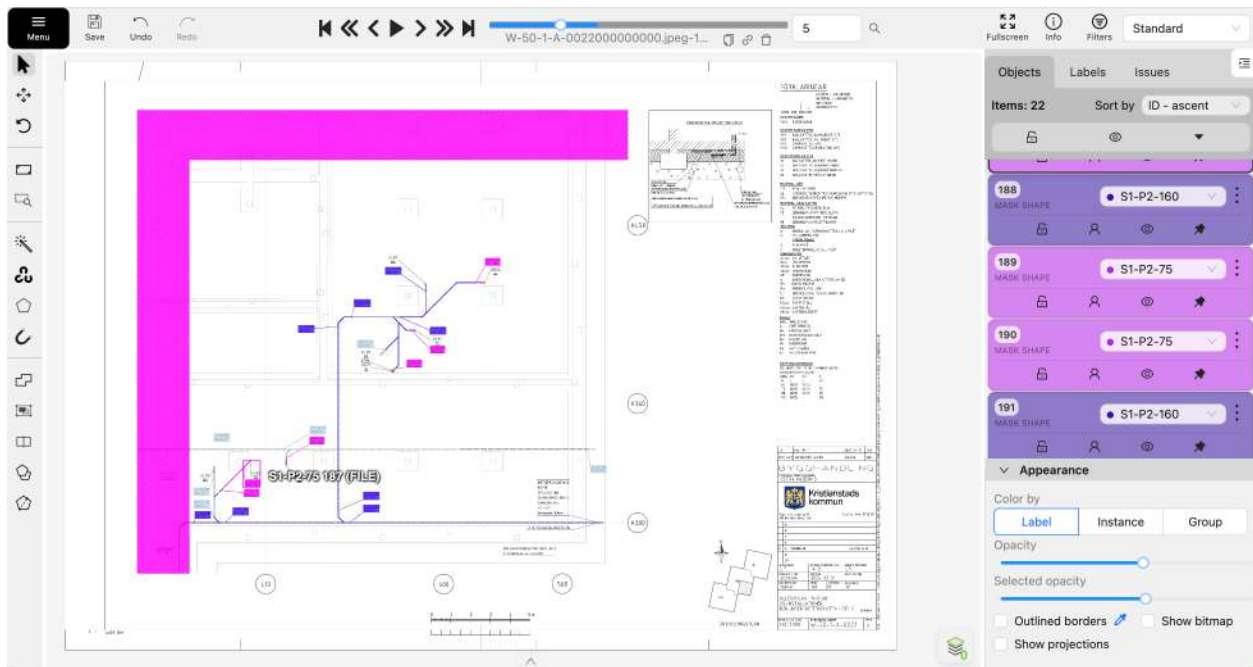


Figure 7.2 [An engineering drawing after annotation]

Example: 8

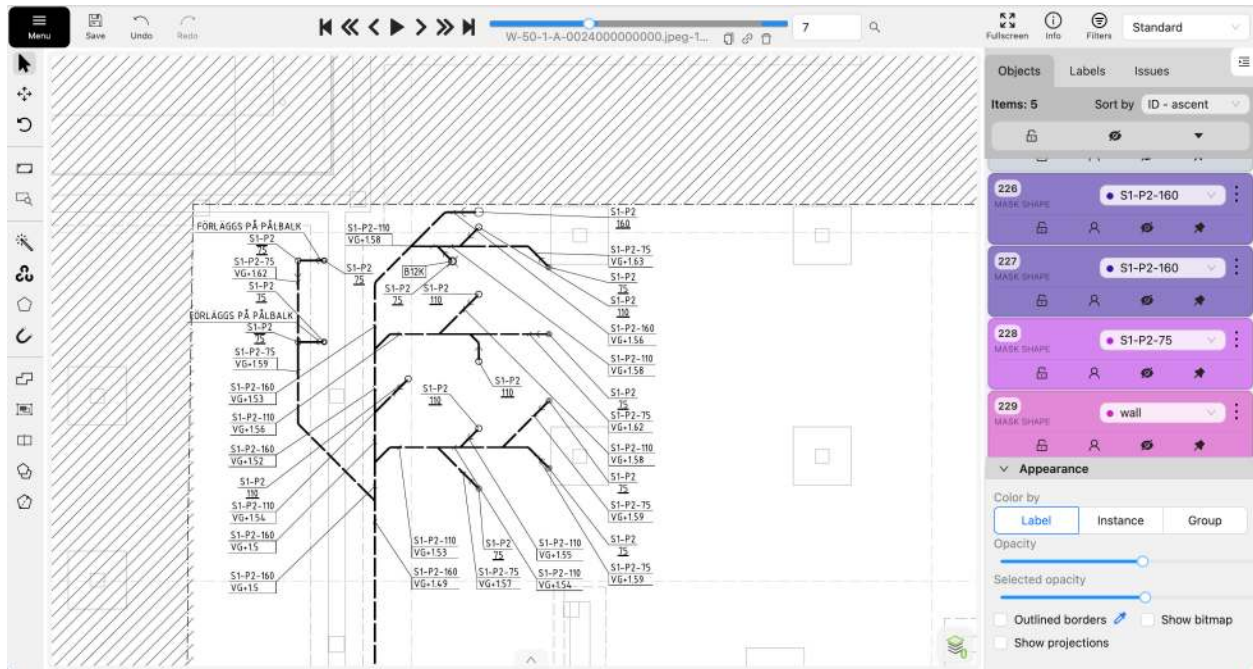


Figure 8.1 [A zoomed engineering drawing before annotation]

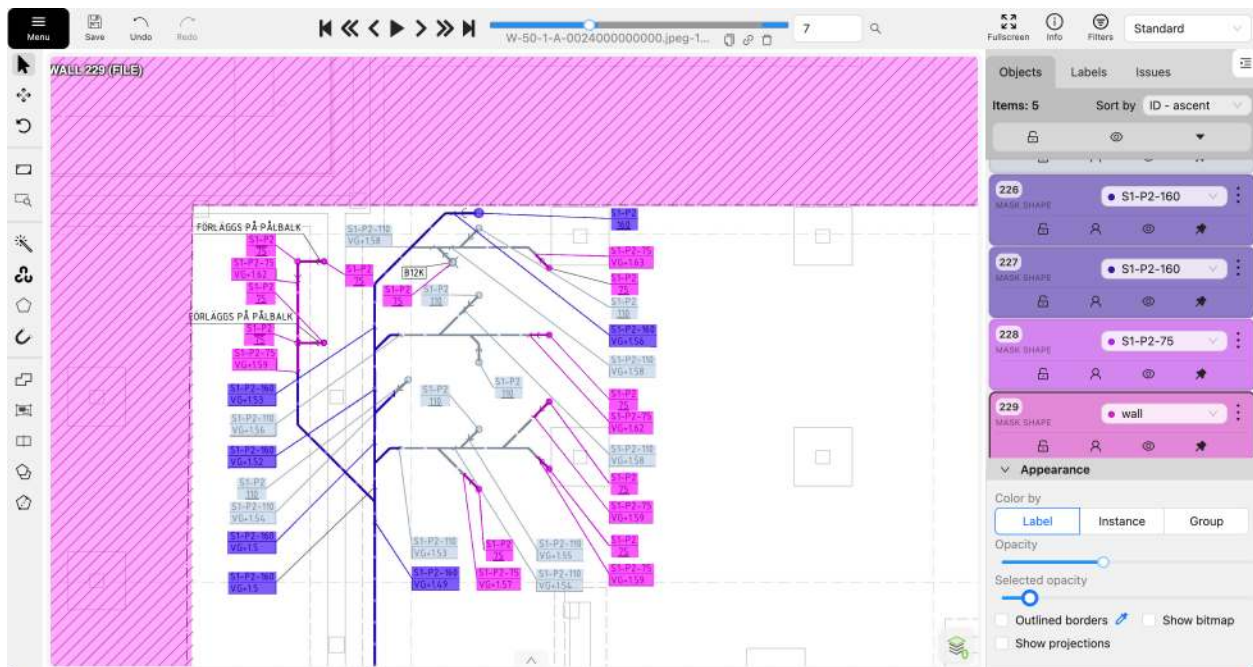


Figure 8.2 [A zoomed engineering drawing after annotation]

Example: 9

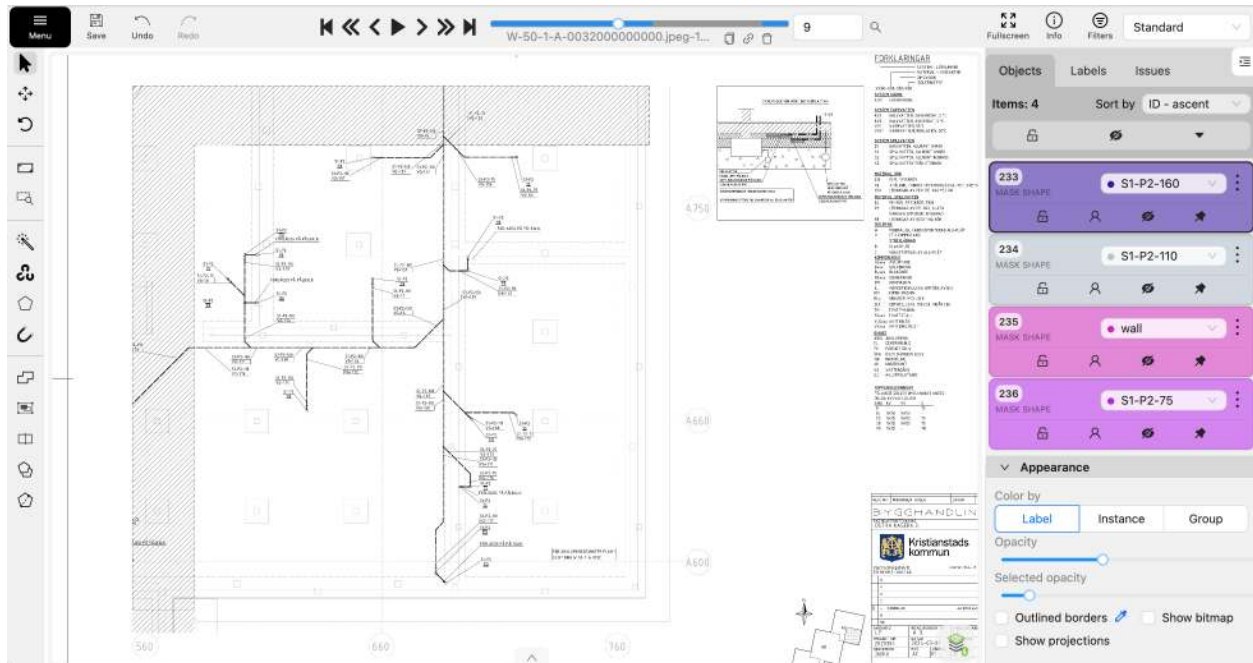


Figure 9.1 [An engineering drawing before annotation]

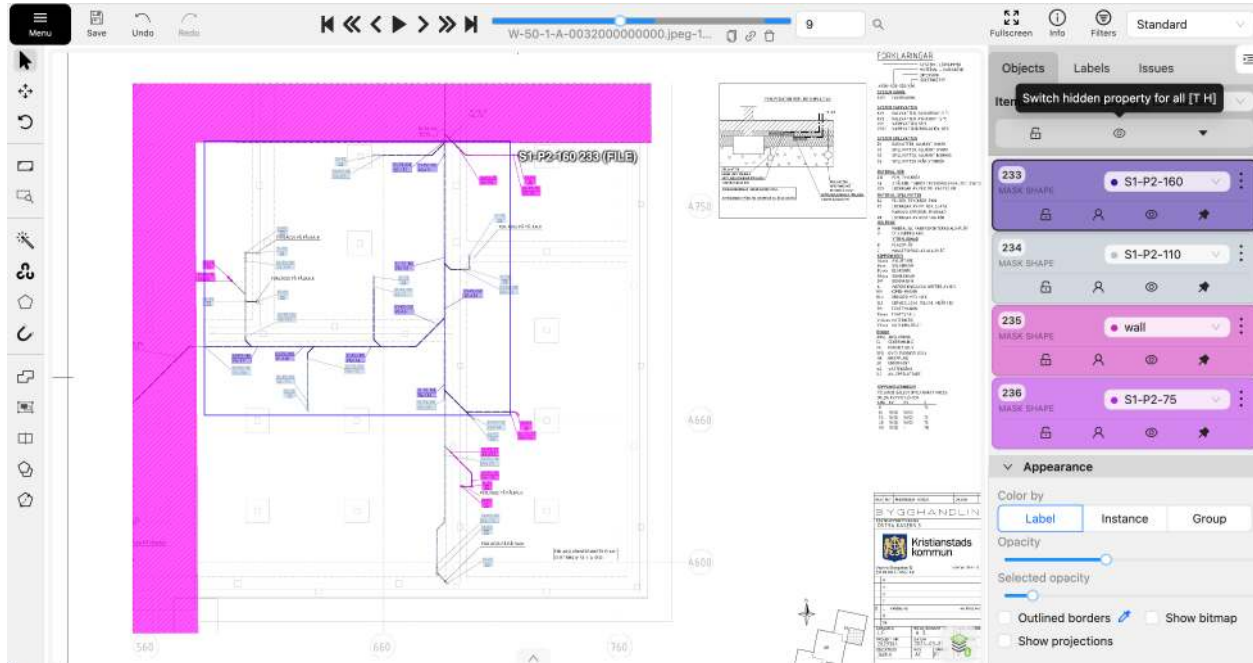


Figure 9.2 [An engineering drawing after annotation]

Example: 10

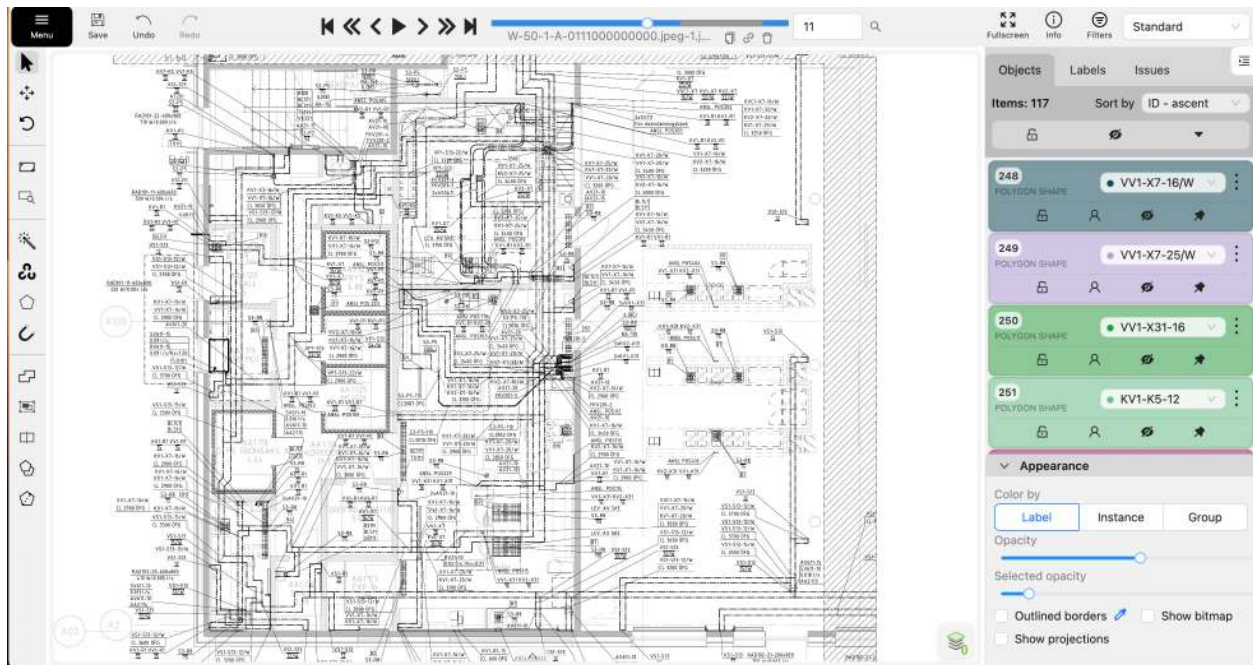


Figure 10.1 [An engineering drawing before annotation]

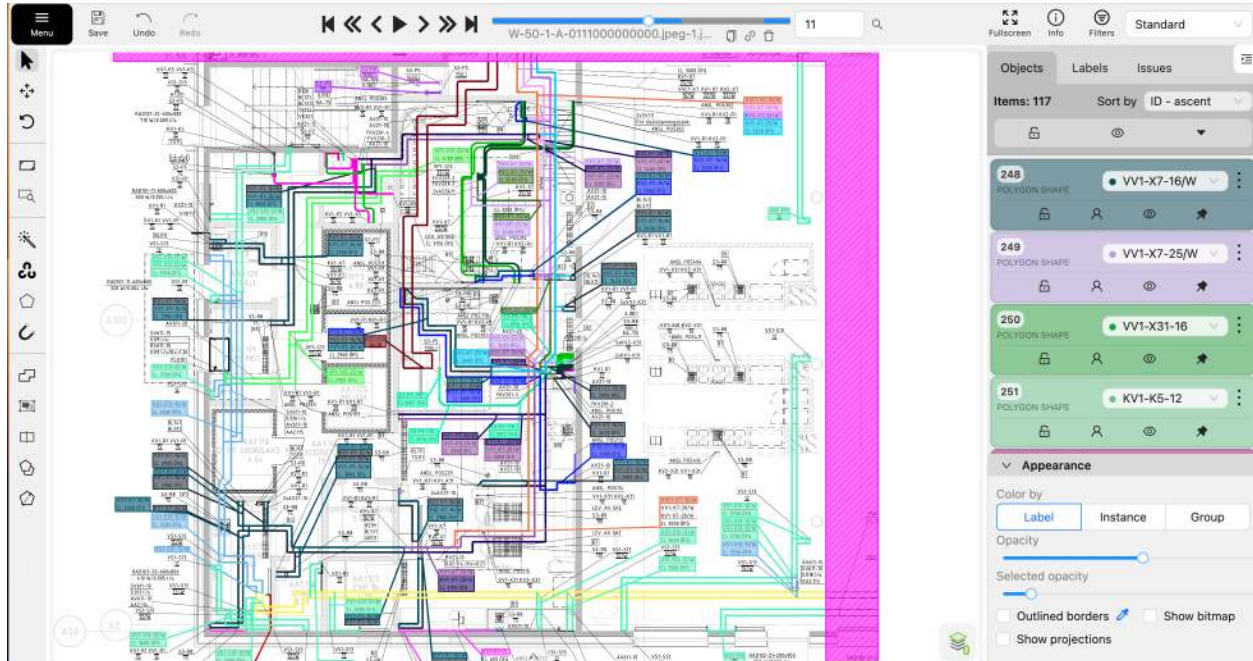


Figure 10.2 [An engineering drawing after annotation]

Example: 11

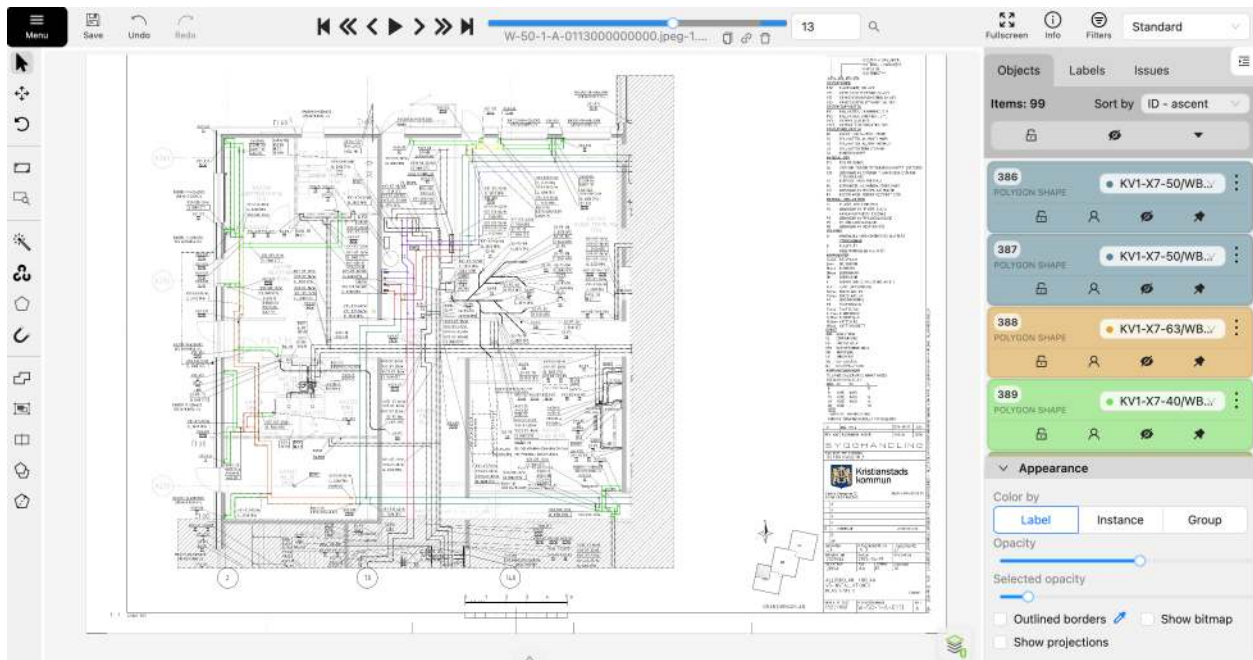


Figure 11.1 [An engineering drawing before annotation]

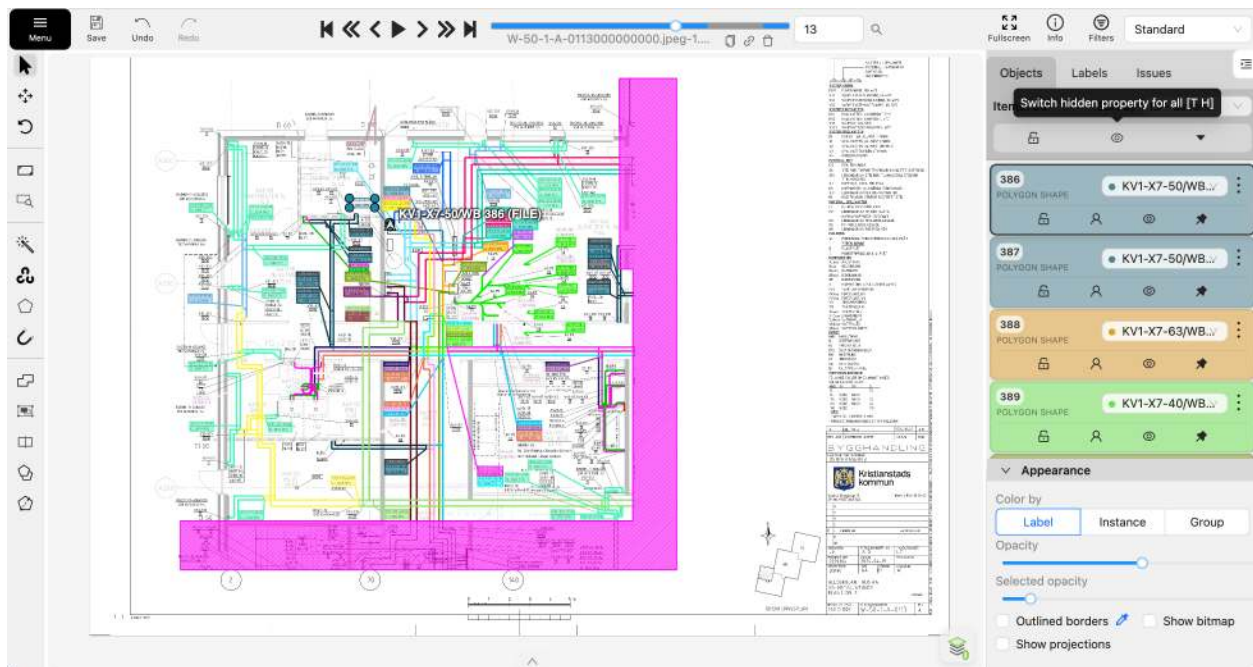


Figure 11.2 [An engineering drawing after annotation]

Example: 12

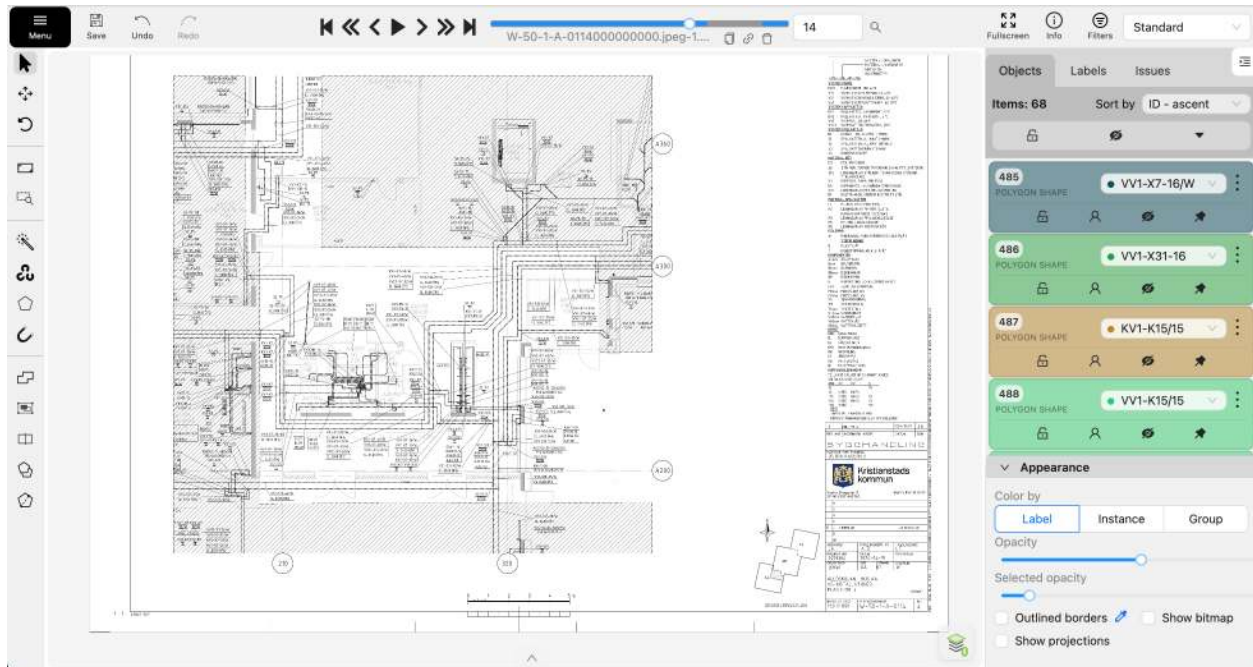


Figure 12.1 [An engineering drawing before annotation]

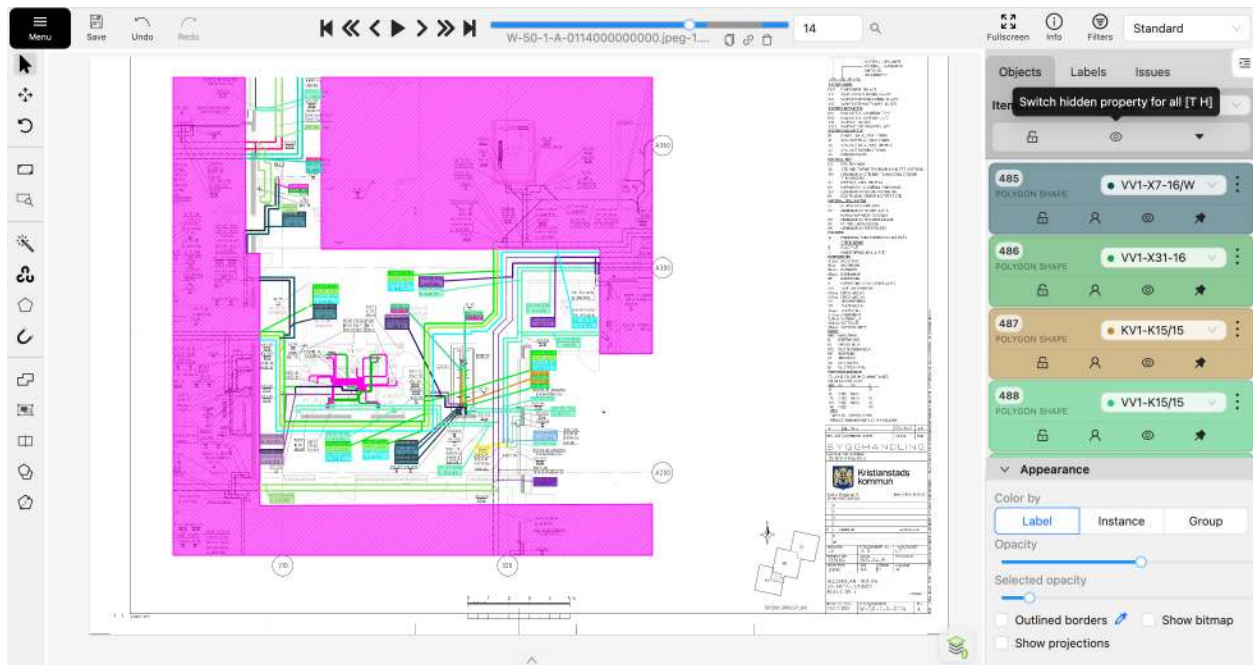


Figure 12.2 [An engineering drawing after annotation]

Example: 13

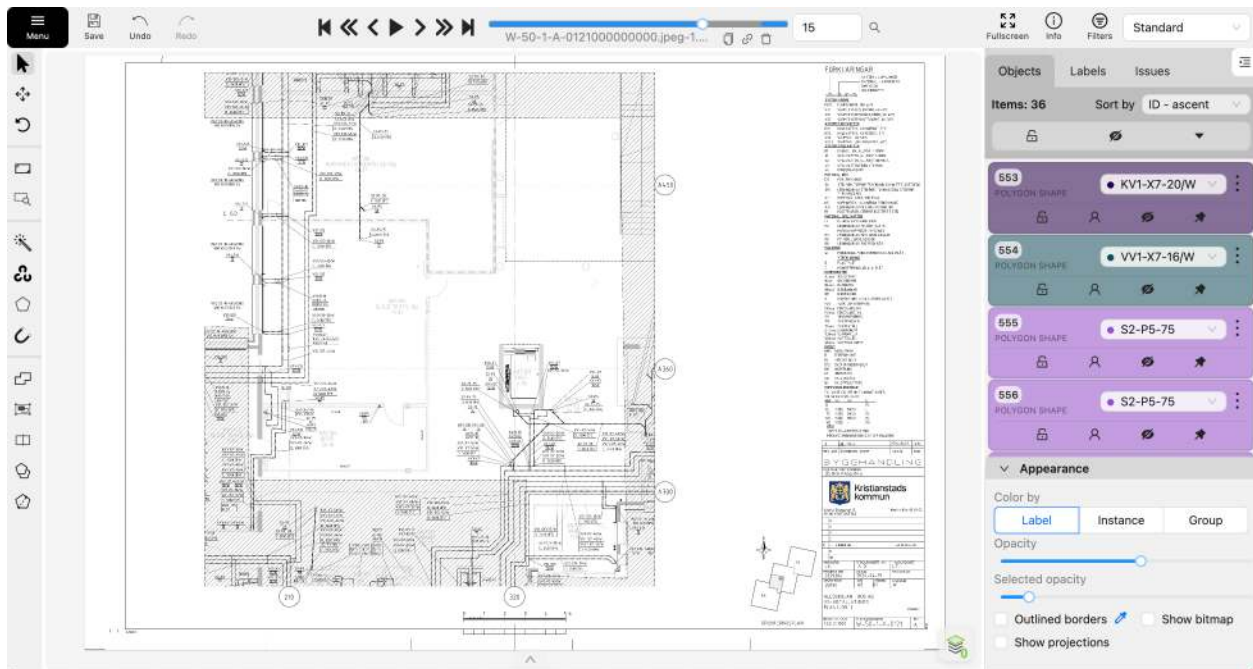


Figure 13.1 [An engineering drawing before annotation]

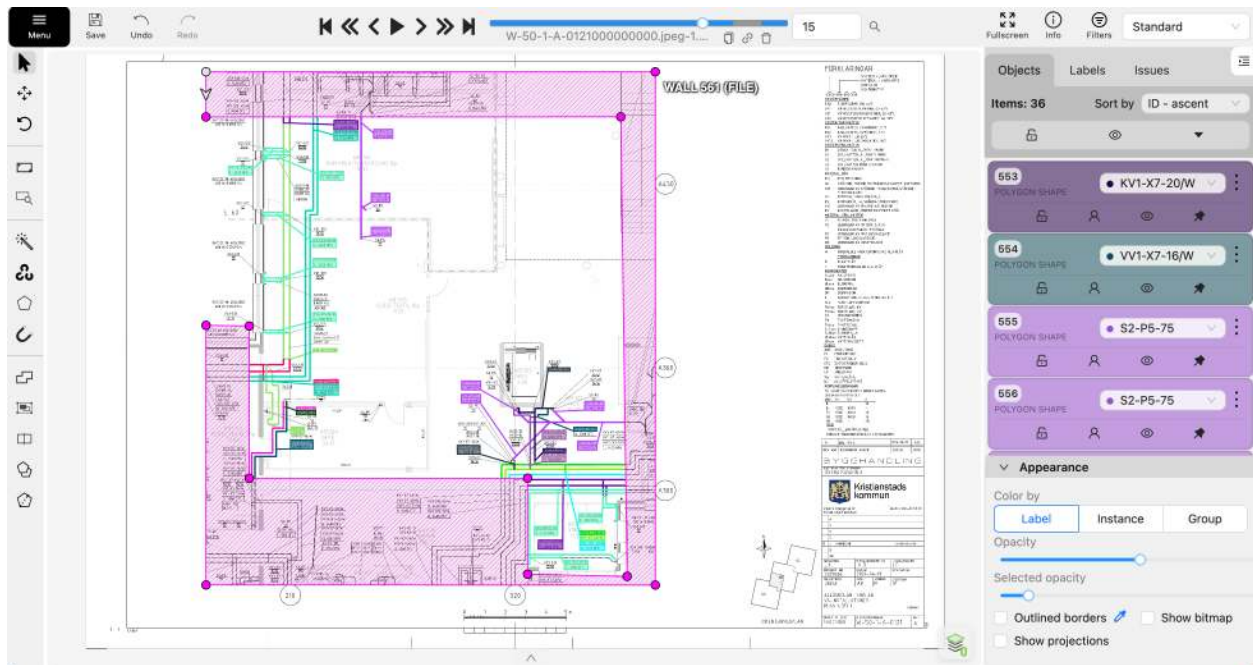


Figure 13.2 [An engineering drawing after annotation]

Example: 14

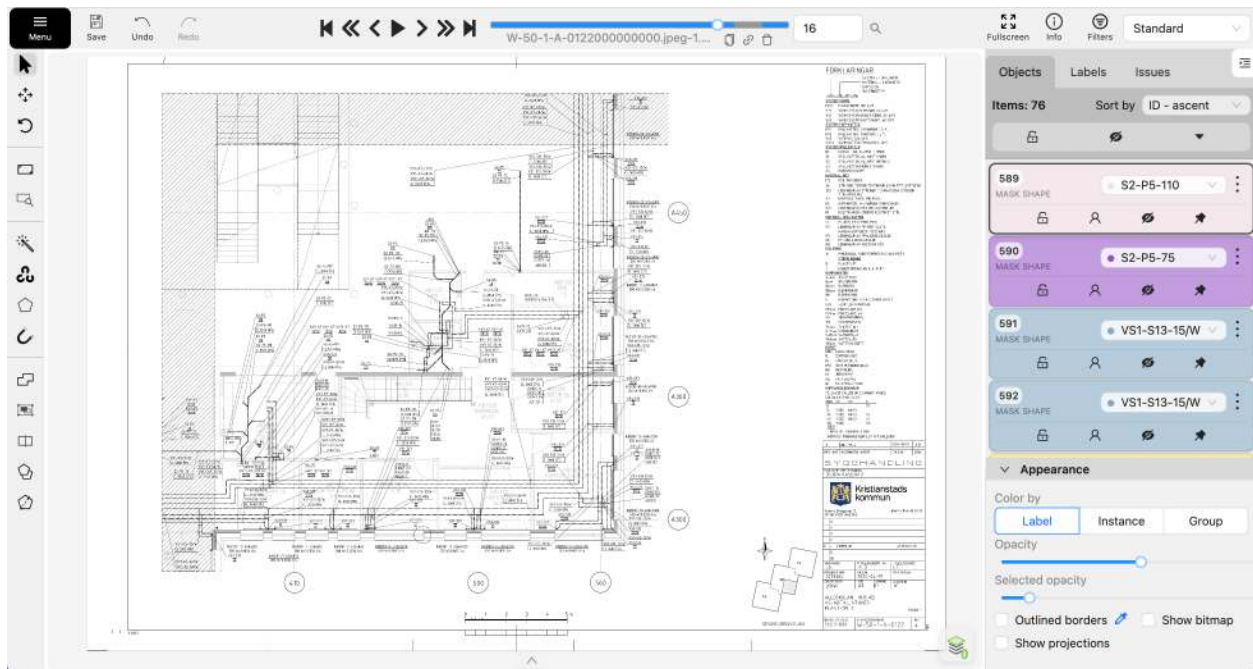


Figure 14.2 [An engineering drawing before annotation]

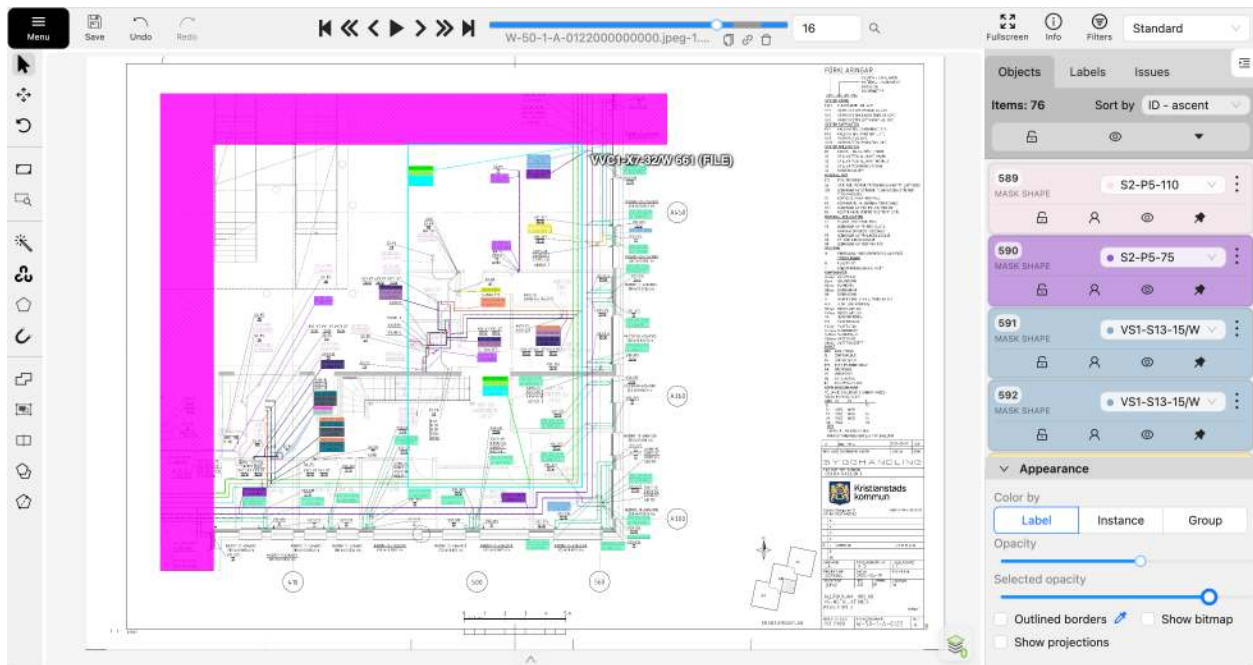


Figure 14.2 [An engineering drawing after annotation]

Example: 16

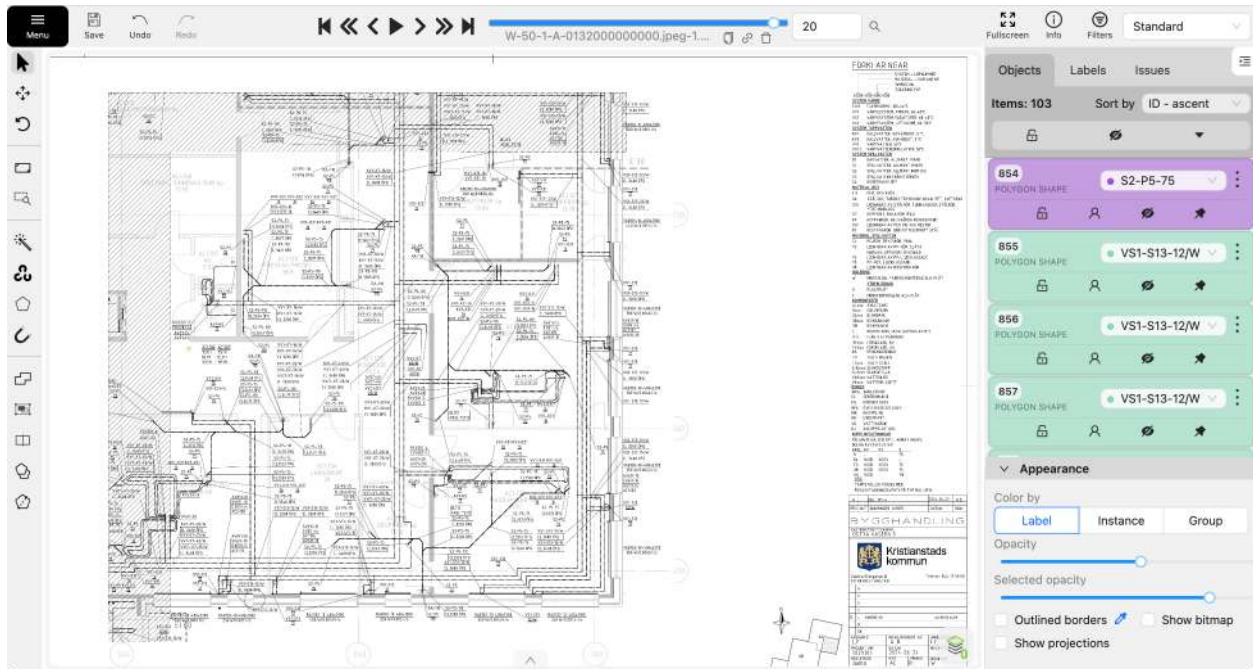


Figure 16.1 [An engineering drawing before annotation]

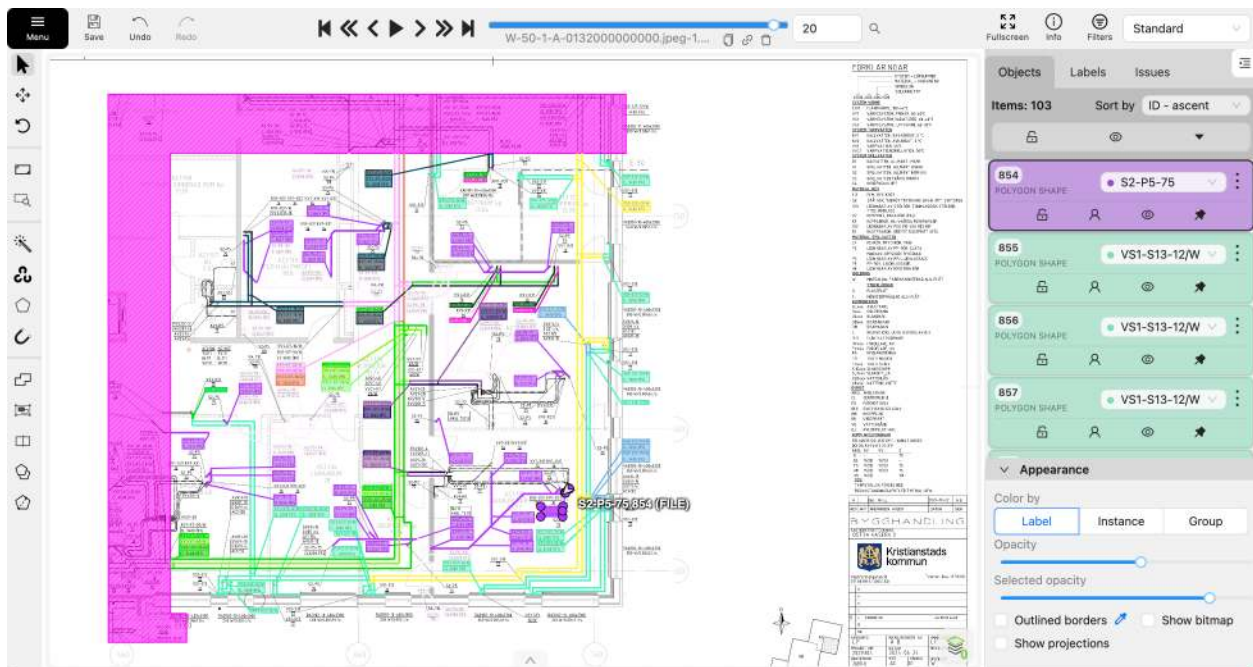


Figure 16.2 [An engineering drawing after annotation]

Example: 17

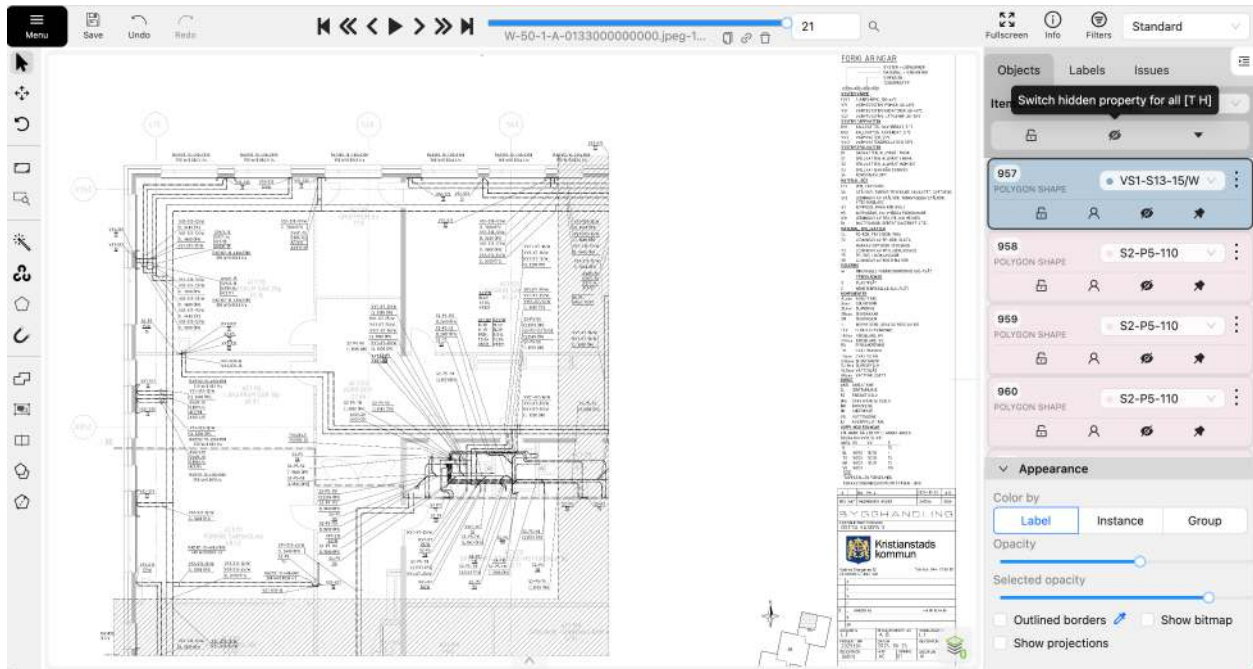


Figure 17.1 [An engineering drawing before annotation]

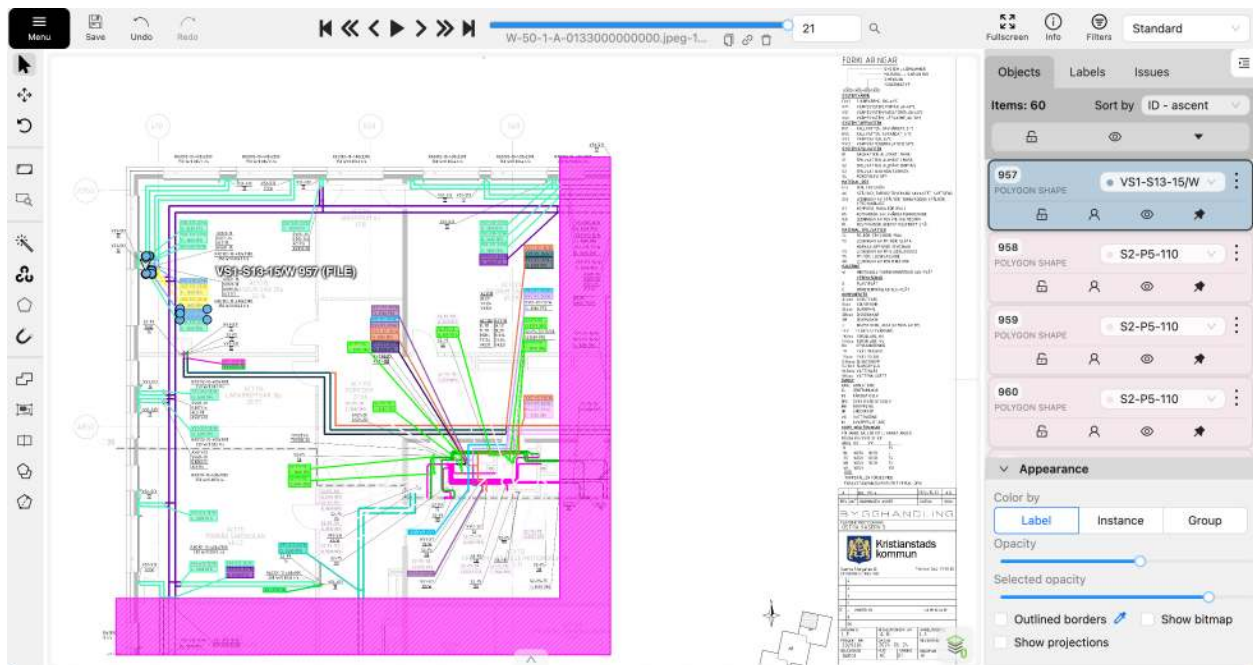


Figure 17.2 [An engineering drawing after annotation]